

Alliance School of Advanced Computing

Bachelor of Computer Applications

ACADEMIC YEAR: 2024-26

Semester – V

Business Analytics

NAME :AARUSH C S

REG NO :2411021240028

MINI PROJECT

MINI PROJECT ANSWER BOOK

Power BI Sales Dashboards

For faculty reference and evaluation only

1. Purpose of this answer book

This book gives the expected solution for each of the six mini projects: the DAX measures, the visuals and the verified figures that a correct dashboard should show. Use it to check student submissions quickly. Designs may differ, but the numbers must match.

2. Projects covered

#	Industry	Table name in Power BI	Main value column
1	Agriculture	Agriculture	Total_Sales_INR

3. Common workflow every student should follow

- **Load:** import the CSV or Excel sheet with Get Data. Set Date to the Date type and every INR, quantity and distance column to Whole or Decimal Number.
- **Clean:** check blanks, duplicates and spelling in Power Query. The datasets are clean, so students should show that they checked.
- **Model:** add calculated columns (Month, MonthNo, Quarter) and at least five DAX measures.
- **Visualize:** build KPI cards, a trend line, a category bar, a donut or treemap, a table and at least two slicers.
- **Interpret:** write three to five insights on the dashboard in a text box.

4. Expected dashboard layout

Area	Content
Top strip	Dashboard title with industry name, plus 3 KPI cards
Left panel	Slicers: Date, main category, one more dimension
Middle left	Line chart of sales by month
Middle right	Bar chart of sales by main category
Bottom left	Donut or treemap of the second dimension
Bottom right	Detail table and text box with insights

Note: *all figures in this book were calculated from the datasets issued to students. Amounts are in INR; Cr = crore (10,000,000) and L = lakh (100,000).*

Project 1: Agriculture Sales Dashboard

Dataset columns: Sale_ID, Date, Crop, Region, Quantity_Quintals, Price_per_Quintal_INR, Total_Sales_INR

A. Key DAX measures (Table: Agriculture)

Measure name	DAX formula
Total Sales	SUM(Agriculture[Total_Sales_INR])
Total Quintals Sold	SUM(Agriculture[Quantity_Quintals])
Total Transactions	COUNTROWS(Agriculture)
Avg Price per Quintal	DIVIDE([Total Sales], [Total Quintals Sold])
Top Crop	MAXX(TOPN(1, VALUES(Agriculture[Crop]), [Total Sales]), Agriculture[Crop])
Bulk sales (over 100 quintals)	CALCULATE([Total Transactions], Agriculture[Quantity_Quintals] > 100)

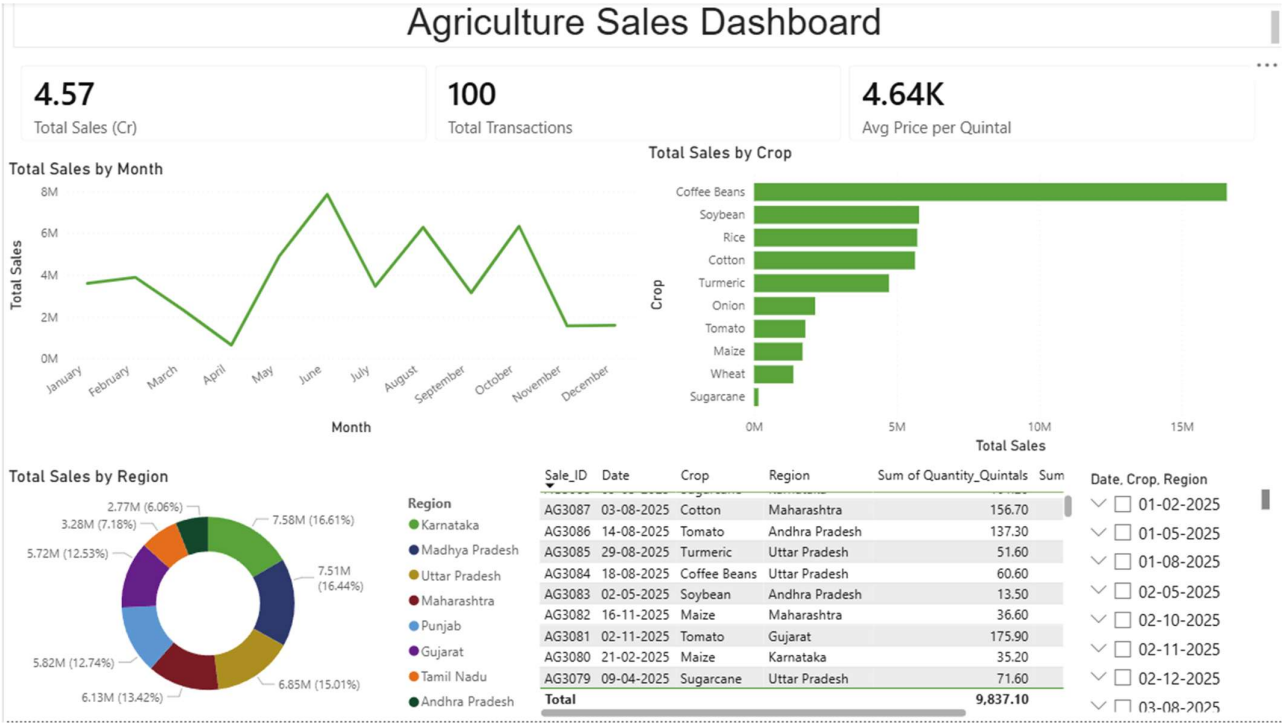
B. Expected visuals

Visual	Fields / settings
KPI cards	Total Sales, Total Transactions, Avg Price per Quintal
Line chart (trend)	Axis: Month (sort by MonthNo) | Values: Total Sales
Clustered bar	Axis: Crop | Values: Total Sales (sort descending)
Donut / Treemap	Legend: Region | Values: Total Sales
Table	Sale_ID, Date, Crop, Region, Quantity_Quintals, Total_Sales_INR
Slicers	Date, Crop, Region

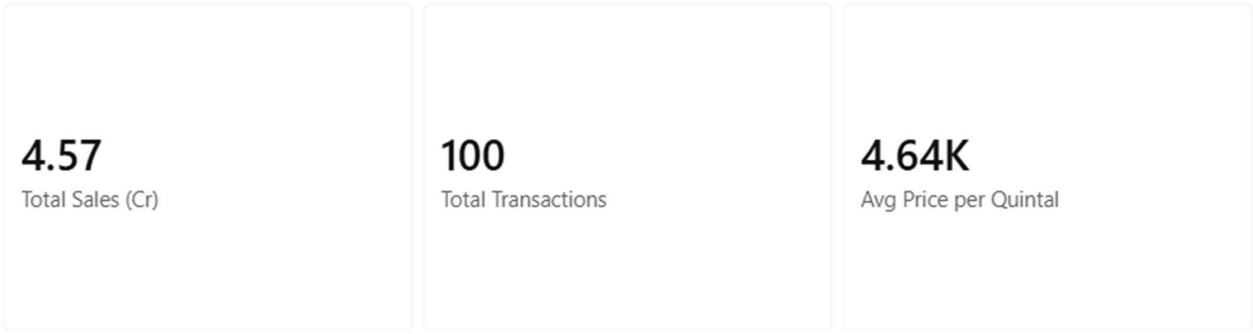
C. Answer key: expected values and insights

- **Check values:** Total Sales = ₹4.57 Cr; Total Transactions = 100; average per record = ₹4.57 L; bulk sales (over 100 quintals) = 43.
- **Insight 1:** Coffee Beans is the top crop with ₹1.66 Cr (36.3% of total); Sugarcane is the lowest at ₹1.54 L.
- **Insight 2:** Karnataka leads by region (₹75.83 L, 16.6%). The best month is June (₹71.13 L) and the weakest is April (₹9.93 L).
- 2 or more slicers and an insights text box.

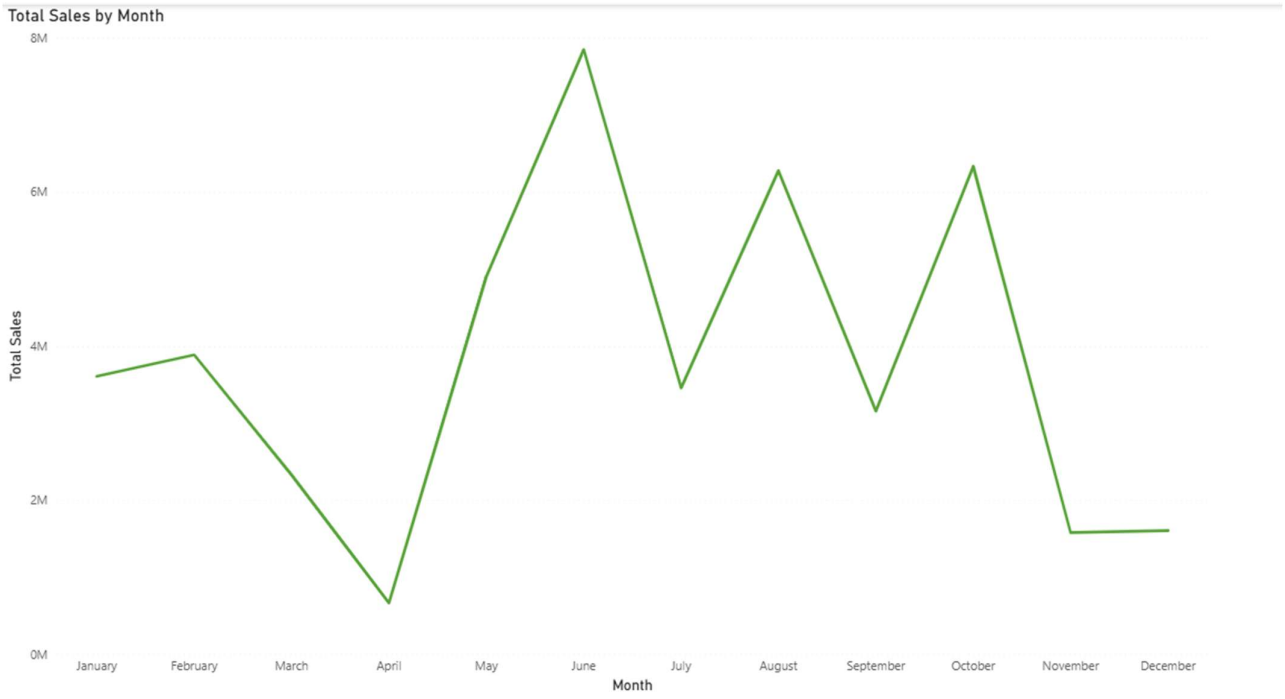
D. Post the Dashboard Screenshots here:



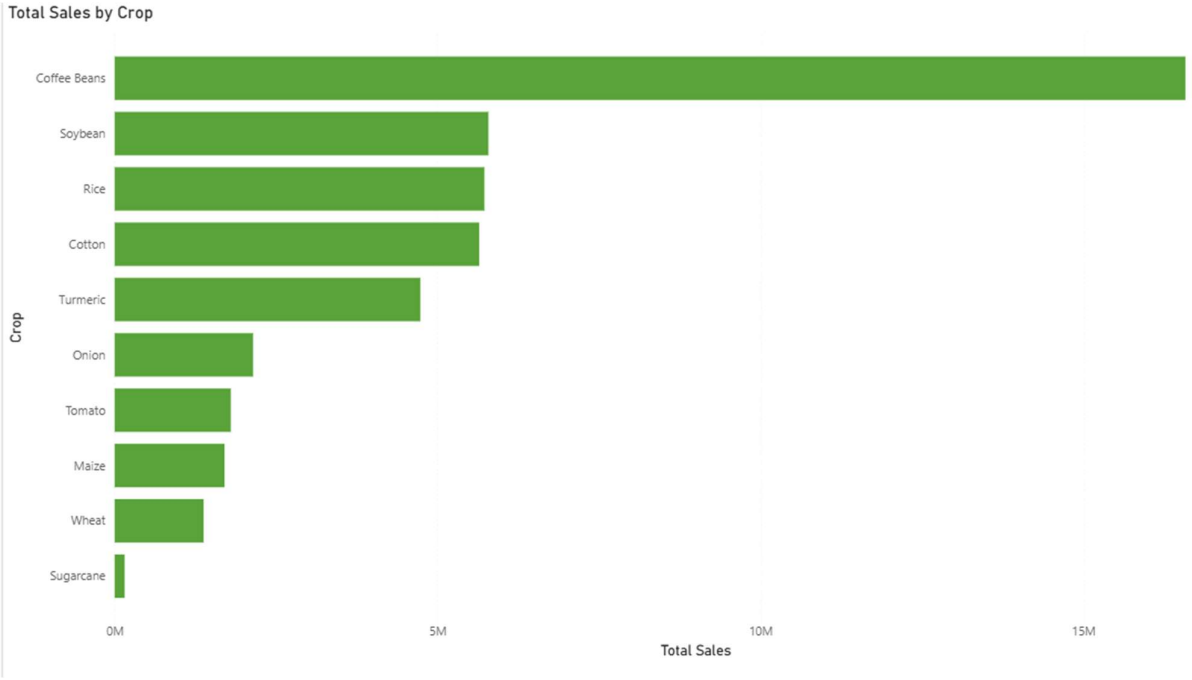
1. KPI cards



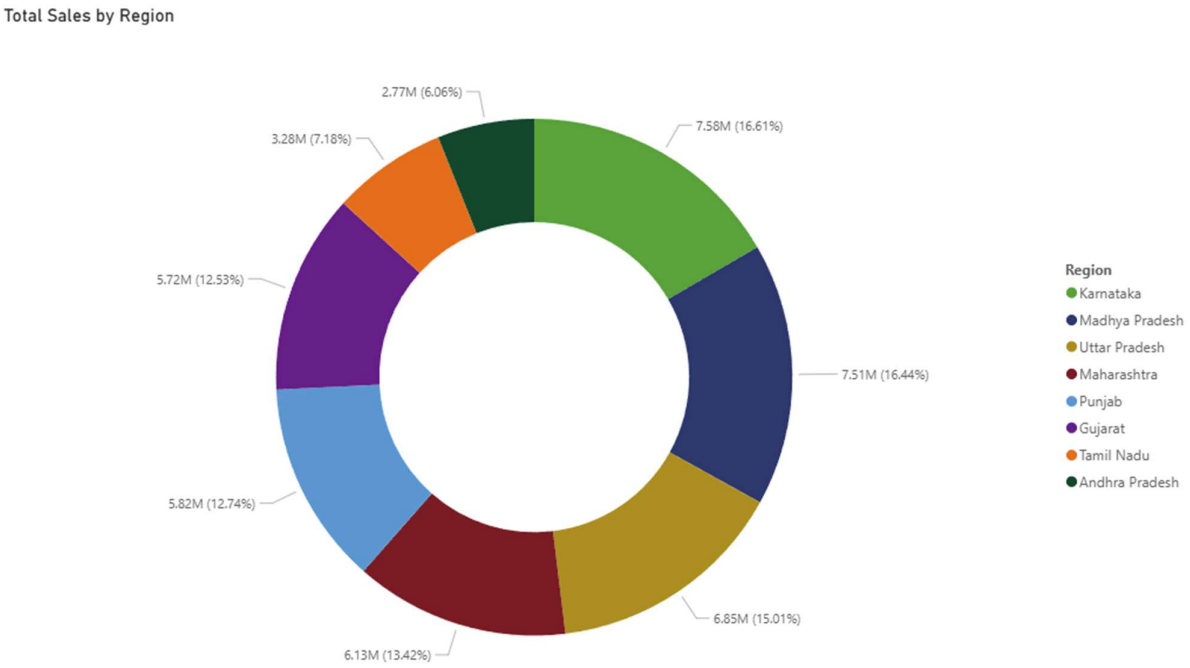
2. Line Chart



3. Clustered Bar chart



4. Donut chart



5. Table

Sale_ID	Date	Crop	Region	Sum of Quantity_Quintals	Sum of Total_Sales_INR
AG3001	09-11-2025	Coffee Beans	Uttar Pradesh	177.40	20,94,207.00
AG3002	25-12-2025	Rice	Tamil Nadu	128.90	3,47,385.50
AG3003	20-01-2025	Onion	Tamil Nadu	27.80	25,576.00
AG3004	03-10-2025	Soybean	Maharashtra	27.60	1,22,406.00
AG3005	23-12-2025	Onion	Gujarat	93.50	2,44,035.00
AG3006	01-08-2025	Coffee Beans	Punjab	149.70	14,63,317.50
AG3007	24-05-2025	Soybean	Karnataka	125.10	6,43,014.00
AG3008	01-05-2025	Tomato	Madhya Pradesh	47.40	84,372.00
AG3009	27-11-2025	Rice	Madhya Pradesh	196.30	5,00,565.00
AG3010	04-03-2025	Wheat	Andhra Pradesh	82.60	1,91,219.00
AG3011	02-11-2025	Rice	Karnataka	134.20	3,97,903.00
AG3012	06-03-2025	Wheat	Uttar Pradesh	158.40	3,71,448.00
AG3013	28-10-2025	Rice	Uttar Pradesh	166.80	4,72,044.00
AG3014	02-11-2025	Sugarcane	Karnataka	69.10	21,766.50
AG3015	08-03-2025	Cotton	Tamil Nadu	172.40	10,71,466.00
AG3016	09-03-2025	Maize	Karnataka	26.40	61,248.00
AG3017	15-06-2025	Tomato	Karnataka	159.00	2,10,675.00
AG3018	04-08-2025	Maize	Punjab	56.70	1,06,312.50
AG3019	20-09-2025	Rice	Punjab	97.90	3,09,853.50
AG3020	23-01-2025	Soybean	Tamil Nadu	103.20	5,30,964.00
AG3021	03-09-2025	Soybean	Gujarat	130.50	5,52,015.00
AG3022	16-08-2025	Onion	Punjab	138.80	2,94,950.00
AG3023	06-03-2025	Rice	Tamil Nadu	67.80	1,50,855.00
AG3024	16-06-2025	Coffee Beans	Gujarat	128.40	15,07,416.00
AG3025	06-11-2025	Coffee Beans	Gujarat	108.50	12,22,795.00
AG3026	02-12-2025	Rice	Uttar Pradesh	159.70	3,46,549.00
AG3027	23-06-2025	Onion	Andhra Pradesh	93.10	1,31,736.50
Total				**9,837.10**	**4,56,52,489.50**

E. Conclusion:

The Agriculture Sales Dashboard was successfully developed using Microsoft Power BI to analyze sales performance across different crops, regions, and months. The dashboard displays key performance indicators, monthly sales trends, crop-wise sales, and regional sales distribution. The analysis shows that Coffee Beans is the highest-selling crop, Karnataka leads in regional sales, and June records the highest monthly sales. Interactive slicers allow users to filter the data by date, crop, and region. Overall, this project demonstrates how Power BI can transform agricultural sales data into meaningful insights that support better business decisions and sales planning.